

JOSE MIGUEL ALDERETE LOPEZ

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PROFESSIONAL SUMMARY

Civil Software Engineer with an IT minor and experience in data engineering, data science, and infrastructure. Designed real-time ETL pipelines, optimized relational databases, and developed explainable machine learning systems, integrating Python, R, SQL, and orchestration tools such as Kubernetes, Terraform, and Ansible.

WORK EXPERIENCE

Data Analytics Intern – Colchones Rosen

Temuco, Chile | Jul 2025 – Oct 2025

- Designed and implemented a real-time ETL pipeline integrating 10,000 to 50,000 daily events from 3 sources (PLC, scanner, and conveyor belt), calculating operational cycle time and efficiency metrics.
- Reduced ingestion pipeline failures by over 70% using state recovery mechanisms and idempotency to handle connectivity loss and duplicate data.
- Optimized SQL queries in MariaDB using CTEs and batch processing, reducing server load during metric calculations.
- Developed analytics modules for daily statistics by operator, product, and process, visualizing KPIs through Grafana dashboards.
- Implemented a PLC state decoder and a fault-tolerant state management system.

FEATURED PROJECTS

ANFIS – Brain Tumor Detection (Bachelor's Thesis)

2023 – 2025

- Designed and implemented a 6-layer ANFIS-Sugeno engine (128 fuzzy rules, 7 GLCM features) for brain tumor classification in MRI images, trained using PSO and least squares.
- Evaluated the model on over 4,000 MRI images, achieving 96.43% accuracy and a 92.15% F1-score on a test sample of 691 images.
- Reduced total pipeline processing time by 33% to 50% through disk caching.
- Built an explainability layer that ranks fuzzy rules by relevance and generates contribution heatmaps.
- Stack:** Python, NumPy, Pandas, SciPy, OpenCV, scikit-image, scikit-learn, scikit-fuzzy, PySwarm, Matplotlib, Seaborn, Tinker.

Kubernetes HA – Automated High Availability Cluster

2024

- Automated the provisioning of a high-availability Kubernetes cluster using Terraform and Ansible, reducing full deployment time to 10–20 minutes.
- Designed a topology with 3 control planes, HAProxy + Keepalived load balancing (single VIP), and distributed storage with Longhorn.
- Achieved control node failover recovery and control in approximately 1 minute.
- Stack:** Terraform, Ansible, Kubernetes, HAProxy, Keepalived, Calico, Longhorn, kubeadm.

Real-Time Manufacturing Analytics Pipeline (AWS)

2026

- Designed a real-time analytics pipeline on AWS (Kinesis, Lambda, DynamoDB, S3, Glue, Athena), ensuring idempotency via DynamoDB transactions and deduplication.
- Modeled infrastructure as code with Terraform (portable to real AWS), featuring least-privilege IAM, SSE-S3 encryption, and a date-partitioned data lake.
- Exposed cycle and quality KPIs in Grafana and automated validation with CI (tests + terraform plan against Floci).
- Stack:** AWS, Terraform, Python, boto3, Grafana, Docker, GitHub Actions.

EDUCATION

Universidad Católica de Temuco – Bachelor of Science in Civil Software Engineering, IT Minor

Mar 2020 – May 2025

CERTIFICATIONS

- EF SET English Certificate – 76/100, C2 Proficient (Advanced English). [\[Verify\]](#)
- ASR Capacitación – Basic Python [\[Verify\]](#) | Intermediate/Advanced Python [\[Verify\]](#)
- ASR Capacitación – Basic Power BI [\[Verify\]](#) | Intermediate Power BI [\[Verify\]](#)
- ASR Capacitación – Introduction to SAP ERP. [\[Verify\]](#)

TECHNICAL SKILLS

Languages: Python, R, SQL, Bash, YAML, HCL (Terraform)

Machine Learning & AI: ANFIS, fuzzy logic, PSO, classification metrics.

Data & Image Processing: NumPy, Pandas, scikit-learn, OpenCV, scikit-image, GLCM.

Visualization: Matplotlib, ggplot2, Grafana, Tableau, Power BI.

Databases: MariaDB, MySQL, CTEs, window functions, query optimization, Redis.

DevOps & Infrastructure: Git, Docker, Ansible, Terraform, Kubernetes, Longhorn, CI/CD.

Cloud & Serverless: AWS (Kinesis, Lambda, DynamoDB, S3, Glue, Athena, IAM), Terraform, boto3, serverless, Floci (AWS emulator).